

On Genomic Foundation Models

Maxime Rochkoulets

Genomic Foundation Models represent a significant part of current research in AI applied to Biology [1–5]. This approach consists in training large deep learning models on huge DNA datasets, using techniques such as autoregressive or masked modeling, essentially training the model to predict unseen tokens, often by leveraging large context sizes.

The goal is that the model will eventually (1) learn some kind of hidden relations or grammar encoded in these sequences (2) develop some kind of general biological understanding such that it can be applied to a variety of downstream tasks. This approach has been very successful in natural language processing, with models such as BERT becoming widely used as a base for downstream tasks fine-tuning.

However, the performance and usefulness of GLMs has been questioned, as it has been observed that specialized solutions (smaller models trained solely on the task, or even classical algorithmic approaches) often outperform GLMs embeddings while being much more convenient to run and use, whereas these models often require extensive resources and compute.

My work here is based mainly on [6], which shows that:

- randomly initialized and smaller models often perform better than large pretrained models on downstream tasks
- large pretrained models embeddings do not show clear advantage over *fully random* embeddings
- large genomic language models are not sensitive to biologically significant changes in sequences

It is worth mentioning that this is not the only paper that demonstrates the poor performance of GLMs, but it is recent and summarizes individual findings of previous papers, so we'll use it as a starting point.

Please note that here I am strictly talking about large models trained on **next-token prediction** from a large corpus of sequences **alone** (and no additional information), with the goal of using their "biologically meaningful" embeddings later, where these *bio-meaningful* embeddings would be a consequence of the model's ability to generalize. This is why for instance I don't consider the fact *they are able to predict the next token much better than random guessing* to be a valid argument against this claim, since it obviously does not entail biological understanding.

Example of those models include Caduceus, DNABERT, Nucleotide Transformer, Evo and Evo2.

This leads me to the following claim

Pretraining on genomic sequences alone by optimizing next-token prediction
is insufficient for models to develop underlying biological understanding.

My main argument is that the success of this approach on natural language is grounded in the distributional hypothesis, of which there is no evidence that it holds for genomic sequences. To me, this implies that we can not expect the same result on nucleic acid sequences simply by applying the same methods.

Additionally, a genomic sequence alone is a very noisy signal (much more than human language), having close to maximal entropy both at the token (bp) and block (k-mer) level [7, 8]. In an information theory paper written by Christoph Adami in 2004 [8] he states that "DNA is a code, and codes do not encode information from sequence alone". By contrast, it seems that mutual information appears when sequences are conditioned on a specific task. This is why much smaller models or more classical approaches outperform in real-world scenarios.

Mans genotypes map to the same phenotype [9], and, to a certain extent, many phenotypes can be induced by the same genotype, making the problem of learning the underlying biological relationships very hard (many-to-many mapping). Sequences with identical function can look different locally and have different next-base statistics, but realize the same phenotype and vice-versa. Intuitively, a model trained on next-token prediction has no reason (pressure, incentive) to map functionally-equivalent-but-sequence-divergent genotypes to nearby points in embedding space.

Of course this is not an issue with genomic sequences only, in natural language, many sentences map to the same meaning or semantics. Nonetheless, LLMs are able to learn complex underlying semantic relationships, and naturally map words

with similar meaning nearby in embedding space... I believe that this ability comes from fundamental differences in the data itself.

The Distributional Hypothesis for natural language states that words that are used and occur in the same contexts tend to have similar meaning.

The DH is a key reason that allows LLMs to implicitly learn semantic structure from next-token prediction. However, I have not been able to find anything that would support such assumption for Genomics, and this in my opinion could be one reason why GLMs do not naturally organize representation space around phenotype.

One more observation, some papers find interesting that models trained from random data also seem to exhibit these biologically meaningful properties [10, 11]. (Or at least properties that authors believe to be meaningful.)

learns to capture overlapping-consistent k-mer embeddings for its token representation within its embedding module. Interestingly, using the k-mer embeddings pre-trained on random data can yield similar performance in downstream tasks, when compared with those using the k-mer embeddings pre-trained on real biological sequences. We further compare the learned k-mer embeddings with other established k-mer represen-

Interestingly, when employing k-mer embeddings of DNABERT pre-trained on random sequences, the DeePromoter model has an improved MCC result compared to the model using k-mer embeddings of DNABERT pre-trained on human data. This improvement is particularly notable in the mouse dataset. In conjunction with the observations from the t-SNE plot shown in Fig. 3, these results suggest that the overlapping-consistent k-mer embeddings learned by DNABERT stand out as a learning outcome for the token representation within the embedding module. Additionally, the k-

overlapping amino acids as their foundational unit. Our assessment across two downstream tasks showed that k-mer embeddings learned from randomized nucleotide sequences can also be helpful in downstream tasks in the pre-training and fine-tuning framework. These embeddings learned from random data can provide close performance comparable to the overlapping k-mer embeddings learned on real genomic data. Furthermore, k-mer embeddings trained on the random data can provide an unbiased option of overlapping k-mer representation for downstream tasks.

Interestingly, in the DNABERT model pre-trained on random DNA 6-mers from Zhang et al.²², we found that the z-score patterns more closely resemble those from the pre-trained model than from the randomly initialized one (Supplementary Figure S3). This suggests that even random DNA sequences can provide enough structure for models to learn generalizable biological signals, consistent with the performance of randomly pre-trained models reported by Zhang et al.

If a model trained on random data is able to exhibit similar properties as the one trained on real sequences, what does that tell us about the initial model, and about the *biologically meaningful*? Especially if we draw a parallel with training an LLM on randomly generated tokens...

References

- [1] Yanrong Ji, Zhihan Zhou, Han Liu, and Ramana V Davuluri. Dnabert: pre-trained bidirectional encoder representations from transformers model for dna-language in genome. *Bioinformatics*, 37(15):2112–2120, 2021.
- [2] Yair Schiff, Chia-Hsiang Kao, Aaron Gokaslan, Tri Dao, Albert Gu, and Volodymyr Kuleshov. Caduceus: Bi-directional equivariant long-range dna sequence modeling. *Proceedings of machine learning research*, 235:43632, 2024.

- [3] Hugo Dalla-Torre, Liam Gonzalez, Javier Mendoza-Revilla, Nicolas Lopez Carranza, Adam Henryk Grzywaczewski, Francesco Oteri, Christian Dallago, Evan Trop, Bernardo P de Almeida, Hassan Sirelkhatim, et al. Nucleotide transformer: building and evaluating robust foundation models for human genomics. *Nature Methods*, 22(2):287–297, 2025.
- [4] Eric Nguyen, Michael Poli, Matthew G Durrant, Brian Kang, Dhruva Katrekar, David B Li, Liam J Bartie, Armin W Thomas, Samuel H King, Garyk Brixi, et al. Sequence modeling and design from molecular to genome scale with evo. *Science*, 386(6723):eado9336, 2024.
- [5] Garyk Brixi, Matthew G Durrant, Jerome Ku, Michael Poli, Greg Brockman, Daniel Chang, Gabriel A Gonzalez, Samuel H King, David B Li, Aditi T Merchant, et al. Genome modeling and design across all domains of life with evo 2. *BioRxiv*, pages 2025–02, 2025.
- [6] Kirill Vishniakov, Karthik Viswanathan, Aleksandr Medvedev, Praveen K Kanithi, Marco AF Pimentel, Ronnie Rajan, and Shadab Khan. Genomic foundationless models: Pretraining does not promise performance. *bioRxiv*, pages 2024–12, 2024.
- [7] Armin O Schmitt and Hanspeter Herzel. Estimating the entropy of dna sequences. *Journal of theoretical biology*, 188(3):369–377, 1997.
- [8] Christoph Adami. Information theory in molecular biology. *Physics of Life Reviews*, 1(1):3–22, 2004.
- [9] Sebastian Edmund Ahnert. Structural properties of genotype–phenotype maps. *Journal of The Royal Society Interface*, 14(132):20170275, 2017.
- [10] Yao-zhong Zhang, Zeheng Bai, and Seiya Imoto. Investigation of the bert model on nucleotide sequences with non-standard pre-training and evaluation of different k-mer embeddings. *Bioinformatics*, 39(10):btad617, 2023.
- [11] Micaela E Consens, Ander Diaz-Navarro, Vivian Chu, Lincoln Stein, Housheng Hansen He, Alan Moses, and Bo Wang. Interpreting attention mechanisms in genomic transformer models: a framework for biological insights. *bioRxiv*, pages 2025–06, 2025.